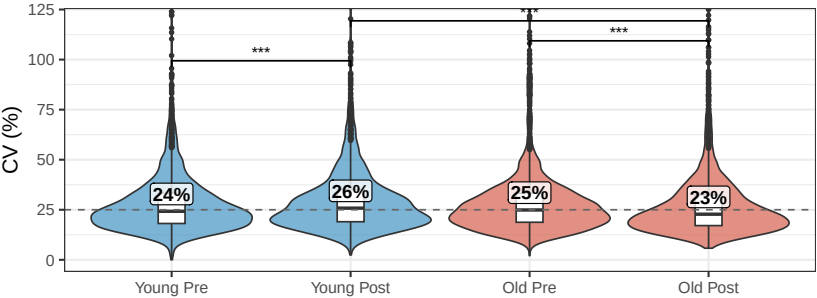


Proteomic Landscape of Aging and Resistance Training Response

Data quality, effect magnitude, contrast overlap, and functional enrichment across the 2 × 2 factorial design.

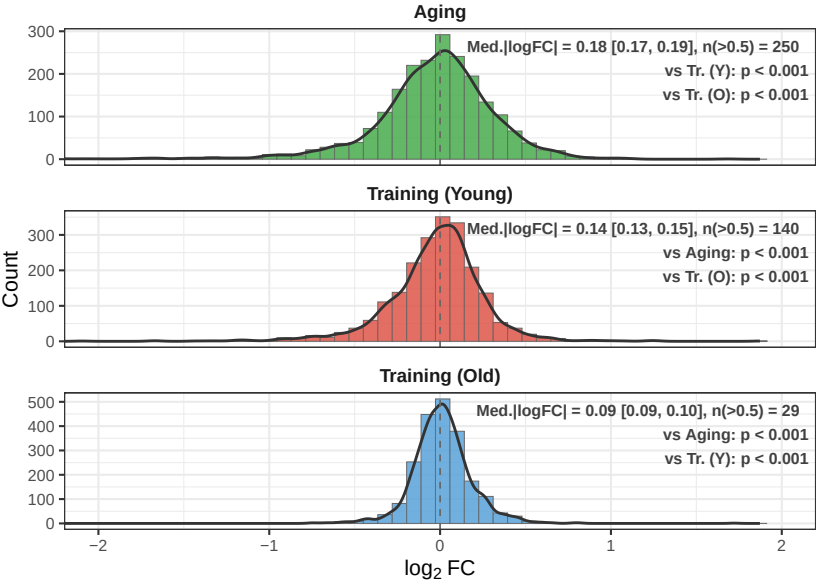
A Inter-Individual Variability (CV%)

Protein-level CV on cycloess-normalized intensities (linear scale)



B Effect Size Distribution (Log2FC)

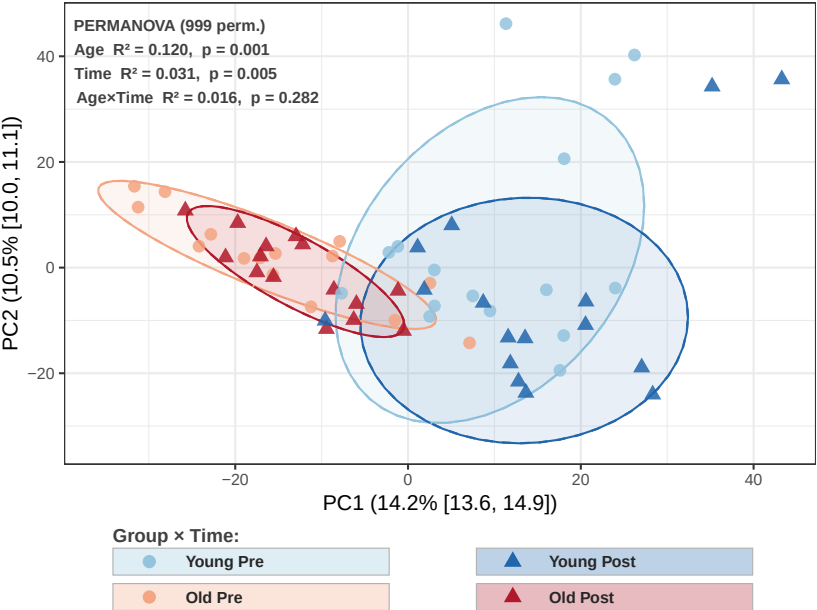
Log2 fold-changes from limma (Interaction excluded)



Young vs Old |logFC|: KS D = 0.18, p < 10⁻⁴ (~30); Fligner .² = 215, p < 10⁻⁴ (~48)
→ Young shows larger, more variable training response

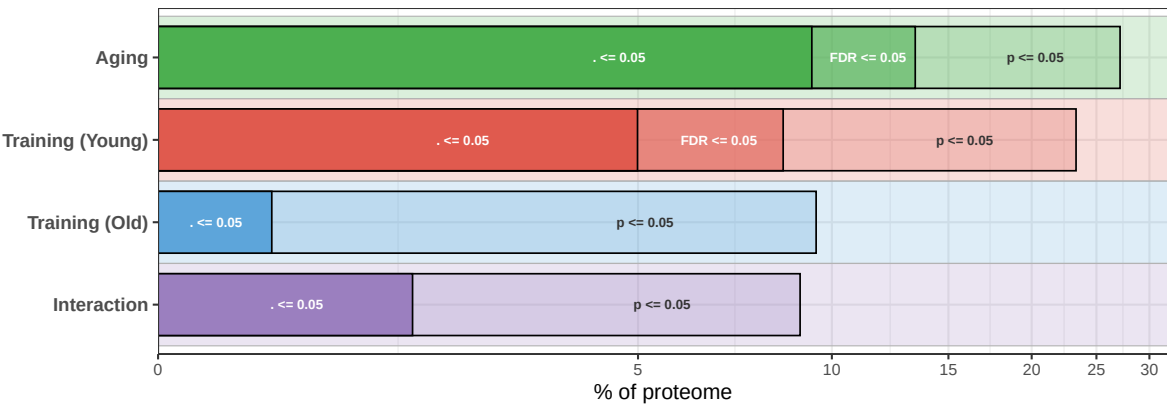
C Principal Component Analysis (PCA)

2,124 proteins (HPA-filtered, cycloess, BPCA-imputed); n = 62 samples



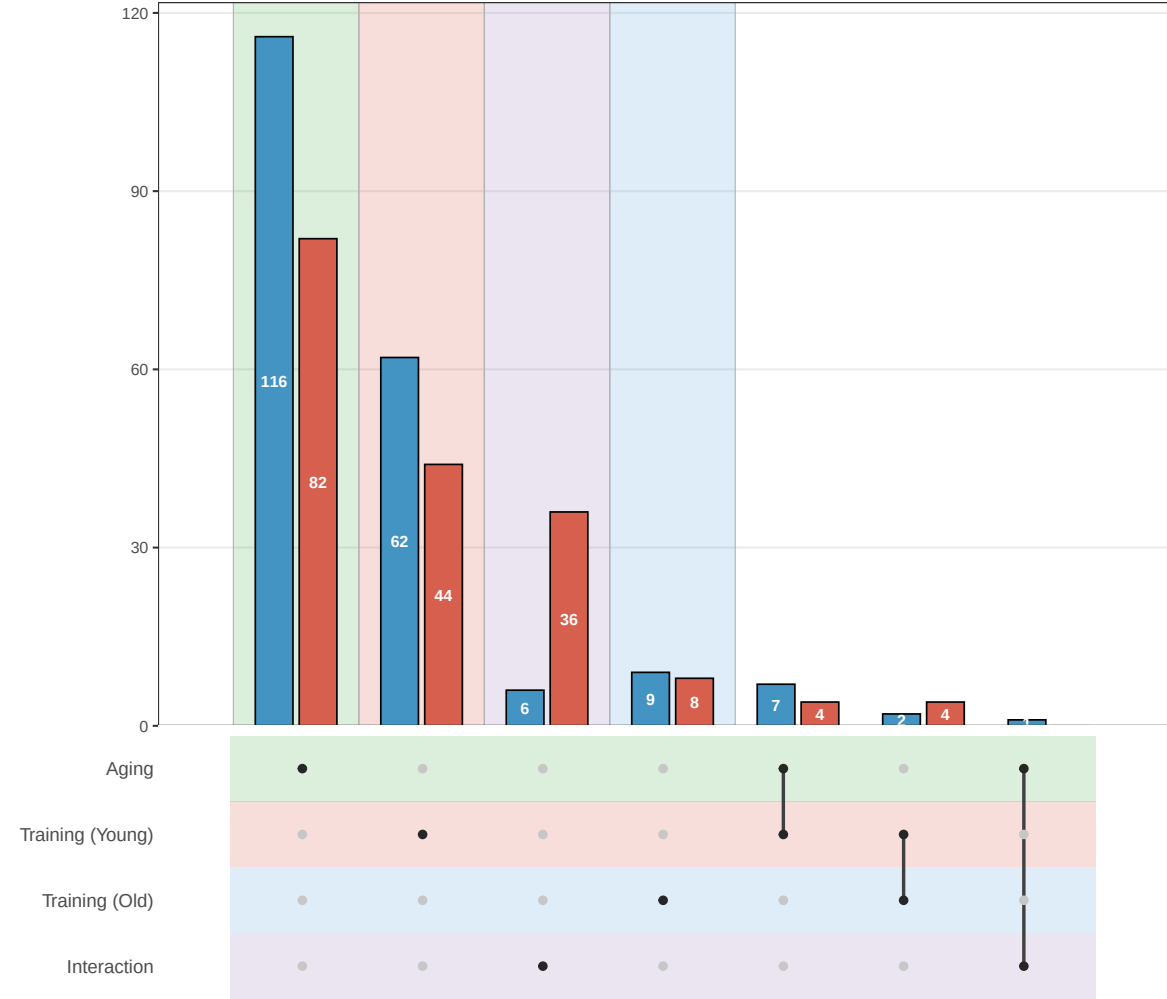
D DEPs per Contrast

Fraction of 2,124 filtered proteins

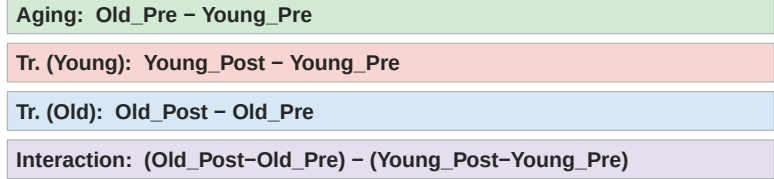


E Contrast Overlap (UpSet)

DEPs by pi < 0.05; bars split by direction | Pi: 363 / FDR: 464 total (59 mixed-direction proteins excluded)



Contrast Definitions:



Direction:
Up
Down

F Pathway Enrichment

fGSEA (padj < 0.05); GO reduced via rrvgo (threshold 0.7)

